

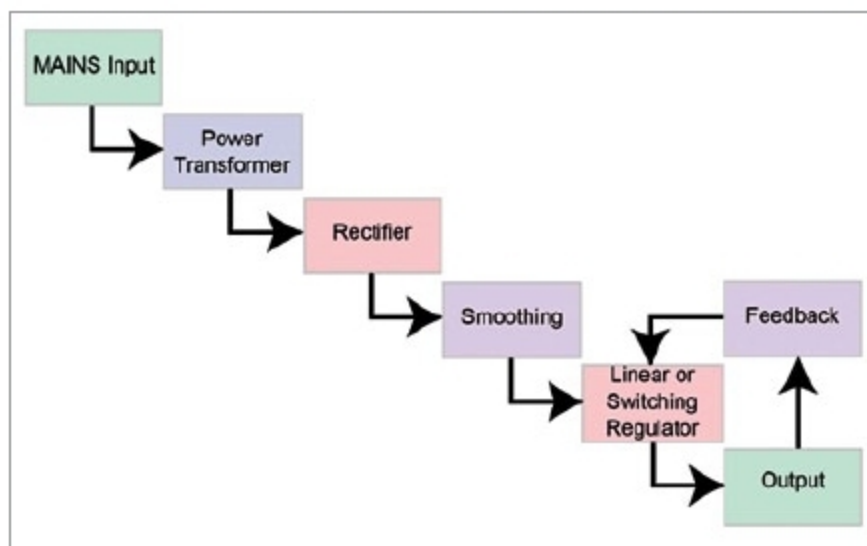


Fig. 1: Block diagram of a regulated supply (Credit: www.teamwavelength.com)

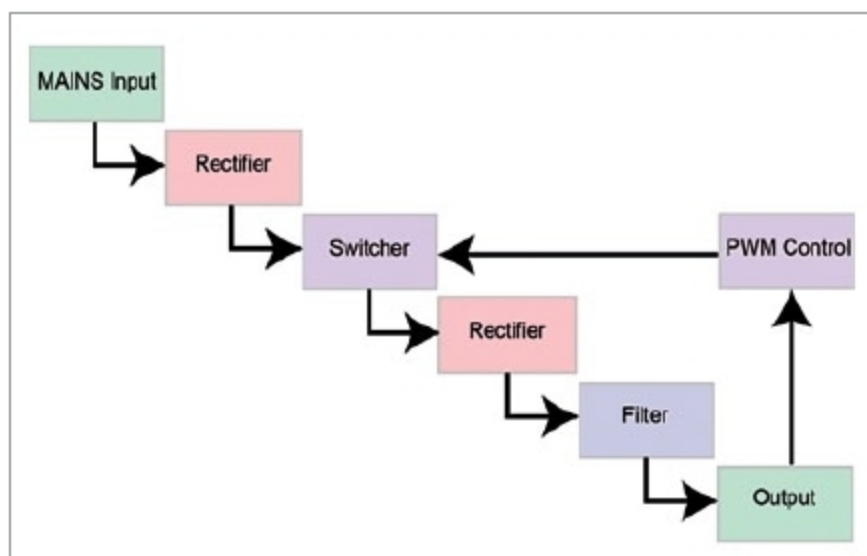


Fig. 2: Block diagram of a regulated switching supply (Credit: www.teamwavelength.com)

that AC components from the power line do not enter at output.

Regulator. This circuit provides constant output virtually, regardless of output current or any minor fluctuations in input level. It may use either linear or switching mode technique to bring output voltage to the required level.

Linear power regulator. This can be used when stepping down of input voltage to regulated output voltage is required. It must be selected based on load current, and input and output voltage requirements.

A linear supply with a low-drop-

There are different topologies available for switching regulators, namely, buck, synchronous buck, boost, inverting buck-boost, SEPIC, Cuk, Zeta, fly-buck, flyback, two-switch flyback, active-clamp forward, single-switch forward, two-switch forward, half-bridge, full-bridge and phase-shifted full bridge. The most popular ones are buck, boost, buck-boost, SEPIC and Zeta.

Boost format steps up the input while buck format steps it down. A buck-boost converter can do both, in addition to inverting the polarity of output voltage with respect to

out (LDO) regulator provides clean DC to meet the need. An LDO is a special variant of a linear regulator, which is used when conversion ratio is high. There are two drop-out voltages: power supply rejection ratio and common-mode rejection ratio. This is the ability of the power supply to reject differential mode noise that is present between positive and negative inputs.

Switching power regulator. Here, transistors operate by switching high frequency between on and off states. An output parameter is controlled by a varying duty cycle, frequency or phase shift of these transitions.

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TABLE I

TYPES OF POWER SUPPLIES AVAILABLE FOR ELECTRONIC DEVICES

AC-DC power supplies	DC-DC converters	AC-AC	DC-AC
- AC adaptor - Front-end power supply - Centralised power supply - AC-DC brick power supply - High-voltage power supply	- Brick DC-DC converter - Non-brick DC-DC converter - Encapsulated DC-DC converter - Bus DC-DC converter - Point-of-load (POL) converter - Power over Ethernet (PoE) power supply - High-voltage DC-DC converter	- Isolation transformer - Variable AC supply - Frequency converter	- Inverter - Generator